

Code to generate Figure 1 of ‘Glucocorticoid receptor collaborates with pioneer factors and AP-1 to execute genome-wide regulation’

Erin M. Wissink

Loading packages and setting paths

```
library(bigWig)
library(dplyr)
library(plyr)
library(ggplot2)
library(knitr)
library(DESeq2)
library(RColorBrewer)
library(pheatmap)
library(gridExtra)
library(wesanderson)

coordinatesPath <- '../Manuscript_data/'
bigwigPath <- '../Manuscript_data/PRO-seq_bigwigs/Raw/'
outputs <- '../figure_outputs/'

gb_coords <- '../figure_outputs/hg38_refseq_NR_gene_bodies.bed'

#getting lists of bigwig files
posbw <- list.files(path = bigwigPath, pattern = '*_plus.bw')
minbw <- list.files(path = bigwigPath, pattern = '*_minus.bw')

samples <- substr(posbw,4,regexpr('_plus.bw',posbw)-1)

gg_options <- theme(axis.text.x = element_text(size = 8),
                    axis.text.y = element_text(size = 8),
                    axis.title.x = element_text(size = 8, colour='black'),
                    axis.title.y = element_text(size = 8, colour='black'),
                    legend.text=element_text(size=8, colour='black'),
                    legend.title = element_text(size = 8, colour='black'),
                    axis.line = element_line(colour = 'black', size = 0.5),
                    axis.ticks = element_line(colour = "black", size = 0.5)) +
  theme_classic()
```

Finding read counts for each gene from raw bigwigs, using the bigWig package.

```
#read in coordinates
gene_data <- read.table(gb_coords,sep=',')
```

```

colnames(gene_data) <- c('chrom','start','end','name','','strand')

gene_data <- gene_data[gene_data$end>gene_data$start,]

#creating columns in dataframe for data
for (i in 1:length(samples)){
  gene_data[,samples[i]] <- NA
}

for (i in 1:length(samples)){
  pbw <- load.bigWig(paste(bigwigPath,posbw[i],sep = ""))
  mbw <- load.bigWig(paste(bigwigPath,minbw[i],sep = ""))
  gene_data[i+6] <- bed6.region.bpQuery.bigWig(bw.plus = pbw, bw.minus = mbw,
    bed6 = gene_data[1:6], op = 'sum', abs.value = TRUE)
}

gene_data$length <- gene_data$end-gene_data$start+1
gene_data_nr <- gene_data[,c(4,7:19)]
gene_data_nr <- dplyr(gene_data_nr,"name",numcolwise(sum))
write.table(gene_data_nr,paste0(outputs,"GR_gene_counts.txt"),quote = FALSE,sep = '\t')

```

Running DESeq2 to find differentially expressed genes

```

row.names(gene_data_nr) <- gene_data_nr$name
geneDESeq <- gene_data_nr[2:13]

DESeq_table <- data.frame(row.names = colnames(geneDESeq))
DESeq_table$cellType <- factor(c(rep('A549',6),rep('U2OS',6)))
DESeq_table$treatment <- factor(rep(c('1','1','0','0','100','100'),2))
DESeq_table$group <- factor(paste0(DESeq_table$cellType,DESeq_table$treatment))

dds <- DESeqDataSetFromMatrix (countData= geneDESeq,
                              colData = DESeq_table, design= ~group)
dds <- estimateSizeFactors(dds)
idx <- rowSums( counts(dds, normalized=TRUE) >= 75 ) >= 3

dds <- dds[idx,]
dds <- DESeq(dds)

A549_wt_1vs0dex <- lfcShrink(dds, contrast = c('group', 'A5491', 'A5490'), type = 'normal')
A549_wt_100vs0dex <- lfcShrink(dds, contrast = c('group', 'A549100', 'A5490'), type = 'normal')
U2OS_wt_1vs0dex <- lfcShrink(dds, contrast = c('group', 'U2OS1', 'U2OS0'), type = 'normal')
U2OS_wt_100vs0dex <- lfcShrink(dds, contrast = c('group', 'U2OS100', 'U2OS0'), type = 'normal')

vsd <- vst(dds, blind = FALSE)

gene_DESeq <- as.data.frame(assay(vsd))

gene_DESeq$baseMean <- A549_wt_1vs0dex$baseMean
gene_DESeq$A549_wt_1vs0dex_log2 <- A549_wt_1vs0dex$log2FoldChange

```

```

gene_DESeq$A549_wt_1vs0dex_fdr <- A549_wt_1vs0dex$padj

gene_DESeq$A549_wt_100vs0dex_log2 <- A549_wt_100vs0dex$log2FoldChange
gene_DESeq$A549_wt_100vs0dex_fdr <- A549_wt_100vs0dex$padj

gene_DESeq$U20S_wt_1vs0dex_log2 <- U20S_wt_1vs0dex$log2FoldChange
gene_DESeq$U20S_wt_1vs0dex_fdr <- U20S_wt_1vs0dex$padj

gene_DESeq$U20S_wt_100vs0dex_log2 <- U20S_wt_100vs0dex$log2FoldChange
gene_DESeq$U20S_wt_100vs0dex_fdr <- U20S_wt_100vs0dex$padj

gene_DESeq$length <- gene_data_nr[idx, 19]

write.table(gene_DESeq, paste0(outputs, 'gene_DESeq_analysis.txt'), quote = F, sep = '\t')

```

##PCA in Supp. Fig. 1

```

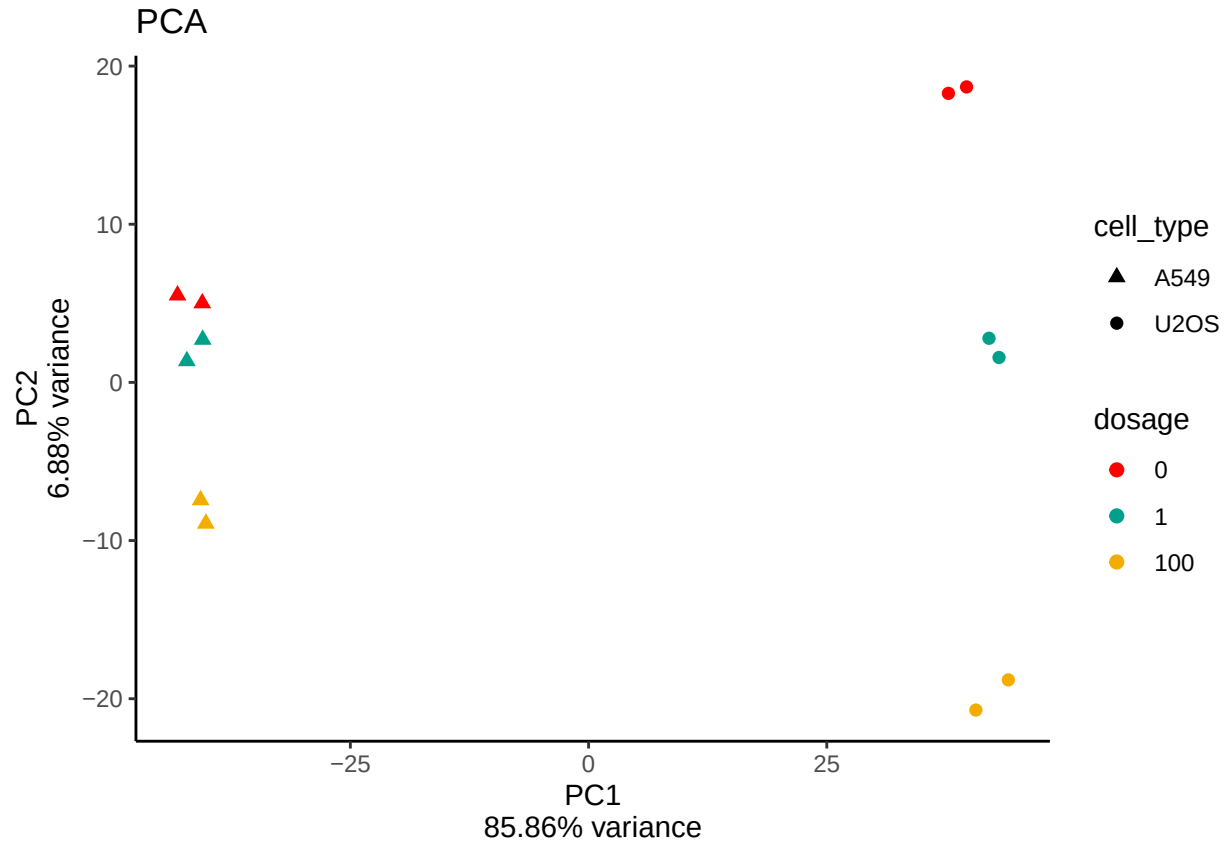
#PCA

forPCA <- t(gene_DESeq[,c(3,4,1,2,5,6,9,10,7,8,11,12)])
pca <- prcomp(forPCA)
summary(pca)

## Importance of components:
##              PC1      PC2      PC3      PC4      PC5      PC6      PC7
## Standard deviation 43.0703 12.3438 6.85136 5.34221 4.54567 3.84802 3.3763
## Proportion of Variance 0.8632 0.0709 0.02184 0.01328 0.00961 0.00689 0.0053
## Cumulative Proportion 0.8632 0.9341 0.95593 0.96921 0.97883 0.98572 0.9910
##              PC8      PC9      PC10     PC11     PC12
## Standard deviation 2.57377 2.37533 2.11284 1.60061 9.296e-14
## Proportion of Variance 0.00308 0.00263 0.00208 0.00119 0.000e+00
## Cumulative Proportion 0.99411 0.99673 0.99881 1.00000 1.000e+00

pca_values <- as.data.frame(pca$x)
pca_values$cell_type <- c(rep(c('A549', 'A549'), 3), rep(c('U20S', 'U20S'), 3))
pca_values$dosage <- rep(c(rep('0', 2), rep('1', 2), rep('100', 2)), 2)
ggplot(pca_values, aes(x = PC1, y = PC2, color = dosage, shape = cell_type)) +
  geom_point(size=2) + ggtitle("PCA") +
  xlab('PC1\n85.86% variance') + ylab('PC2\n6.88% variance') +
  scale_shape_manual(values=c(17,16)) +
  scale_color_manual(values = wes_palette(n=3, name = "Darjeeling1")) + gg_options

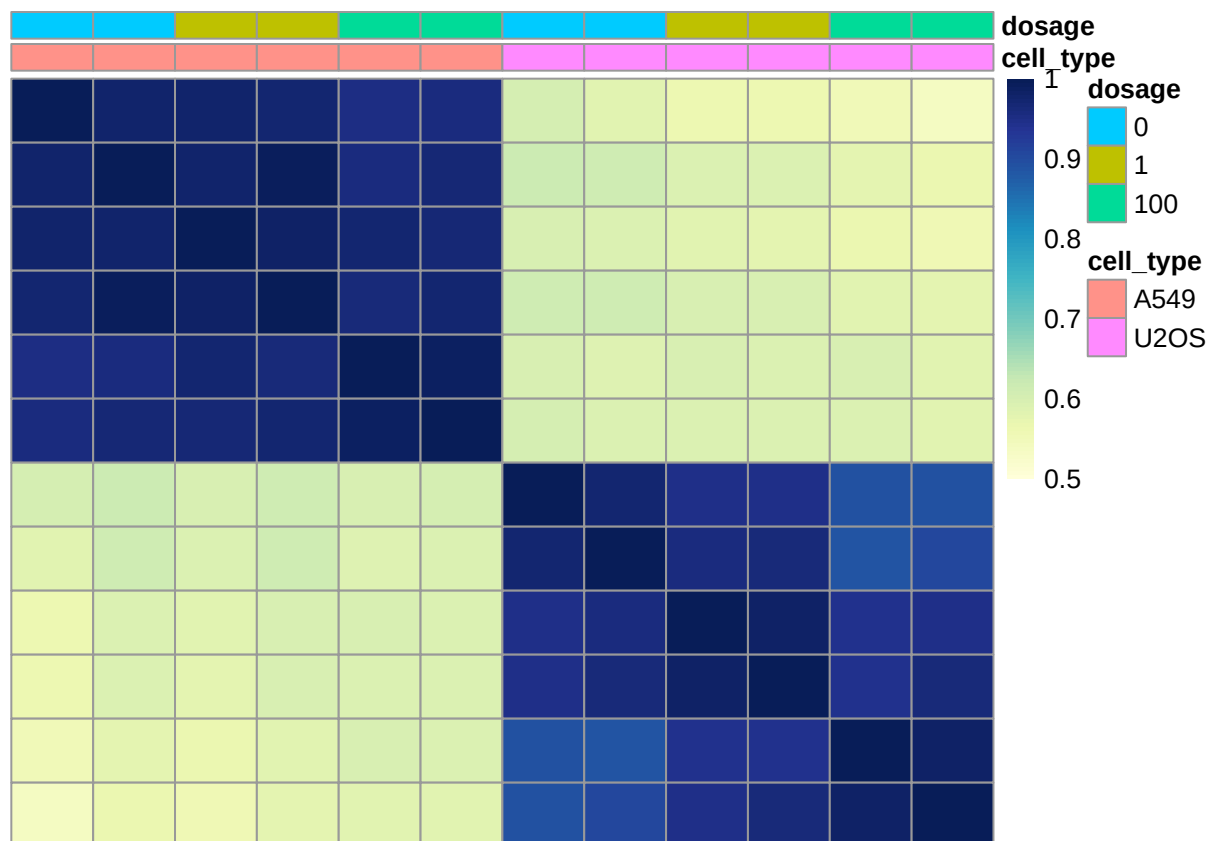
```



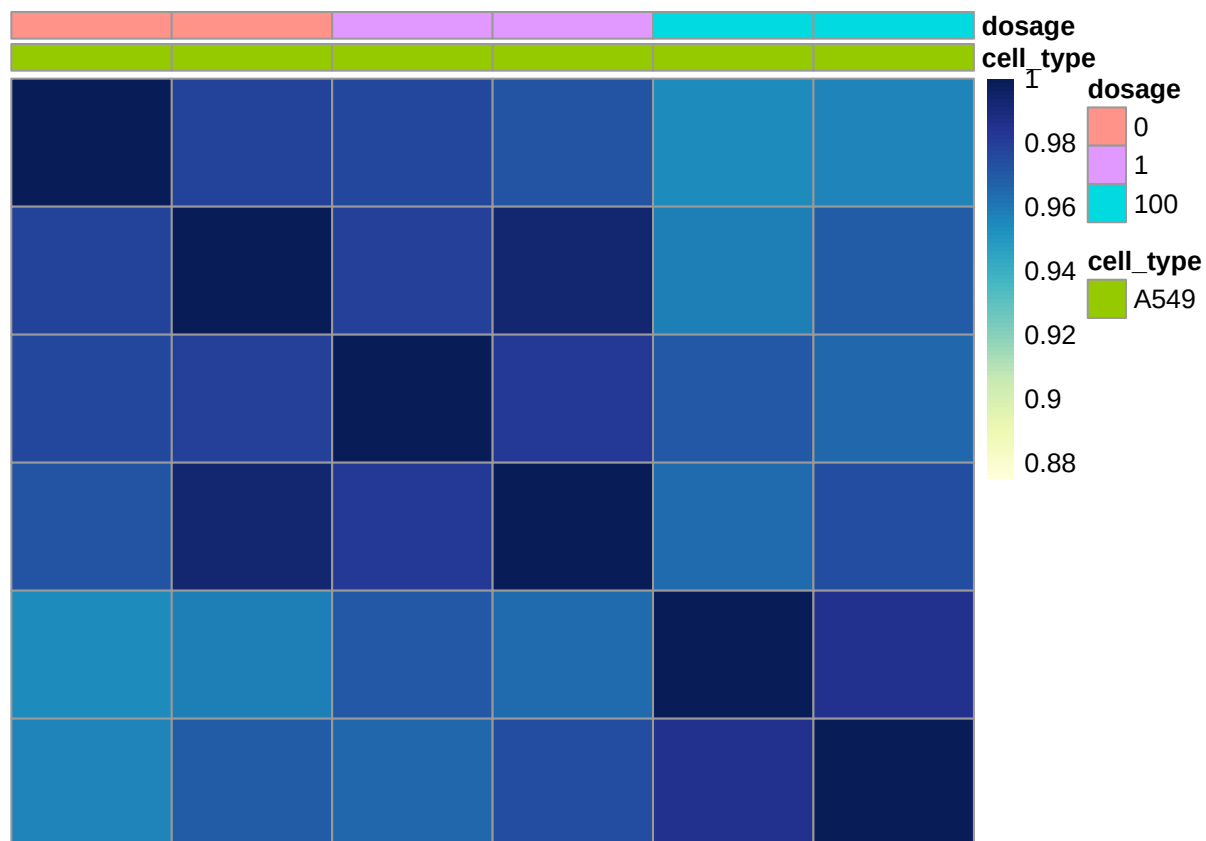
##Spearman correlation plots in Supp. Fig. 1

```
# correlation matrix
corr_matrix <- as.data.frame(matrix(nrow = 12, ncol = 12))
rownames(corr_matrix) <- rownames(pca_values)
colnames(corr_matrix) <- rownames(pca_values)

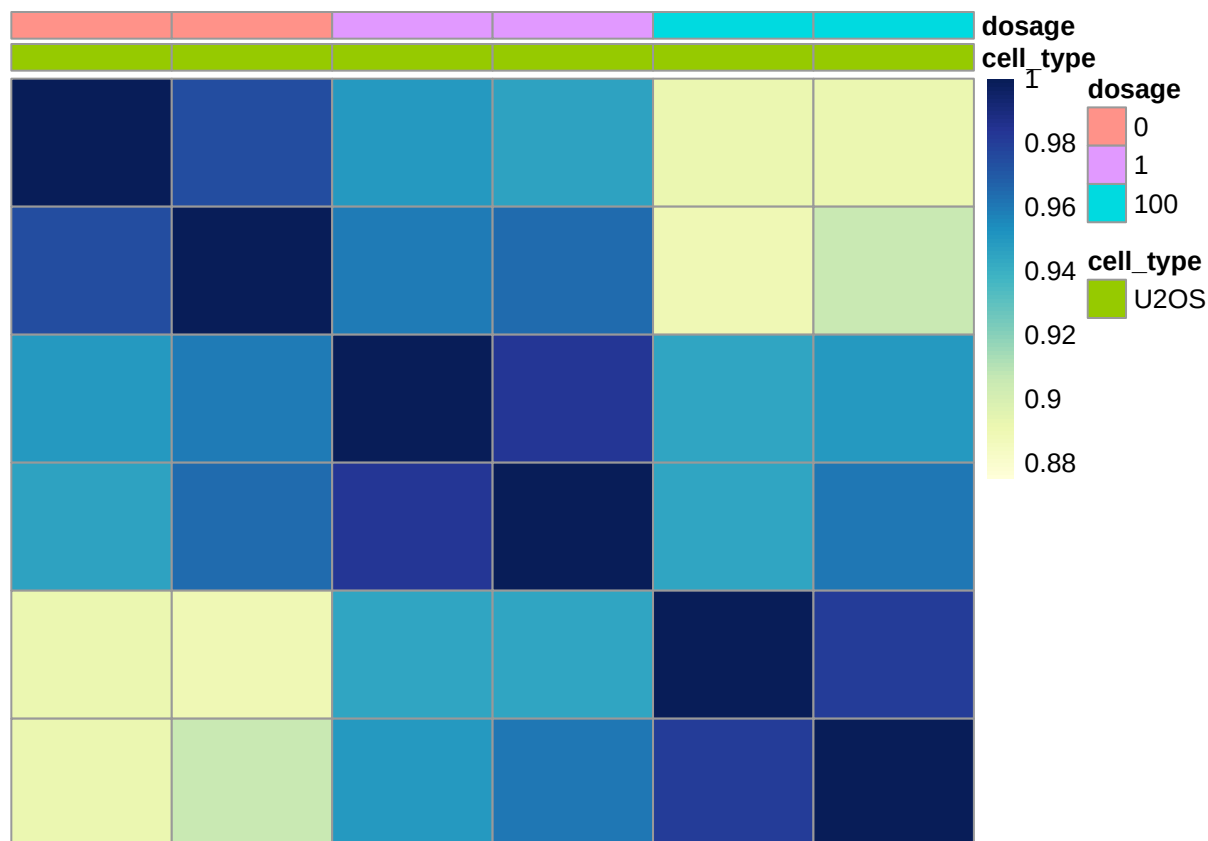
toUse <- c(3,4,1,2,5,6,9,10,7,8,11,12)
for (i in 1:length(toUse)){
  for (j in 1:length(toUse)){
    corr_matrix[i,j] <- cor.test(gene_DESeq[,toUse[i]], gene_DESeq[,toUse[j]],
                                method = 'spearman')$estimate
  }
}
pheatmap(corr_matrix,
          color = colorRampPalette(brewer.pal(n = 9, name = "YlGnBu")) (100),
          breaks = seq(0.5, 1.0, length.out = 101),
          cluster_rows = F, cluster_cols = F,
          annotation_col = pca_values[13:14],
          show_rownames = F, show_colnames = F)
```



```
pheatmap(corr_matrix[1:6,1:6],
  color = colorRampPalette(brewer.pal(n = 9, name = "YlGnBu")) (100),
  breaks = seq(0.875, 1.0, length.out = 101),
  cluster_rows = F, cluster_cols = F,
  annotation_col = pca_values[1:6,13:14],
  show_rownames = F, show_colnames = F)
```



```
pheatmap(corr_matrix[7:12,7:12],
  color = colorRampPalette(brewer.pal(n = 9, name = "YlGnBu")) (100),
  breaks = seq(0.875, 1.0, length.out = 101),
  cluster_rows = F, cluster_cols = F,
  annotation_col = pca_values[7:12,13:14],
  show_rownames = F, show_colnames = F)
```



MA plots and comparison plot for Fig. 1 C-E

```
gene_DESeq[is.na(gene_DESeq)] <- 1
gene_DESeq$sig_1nM <- ''
gene_DESeq$sig_100nM <- ''

for (i in 1:nrow(gene_DESeq)){
  if (gene_DESeq[i,15] < 0.05 & gene_DESeq[i,19] < 0.05){
    gene_DESeq[i,22] <- 'both'
  } else if (gene_DESeq[i,15] < 0.05){
    gene_DESeq[i,22] <- 'A549'
  } else if (gene_DESeq[i,19] < 0.05){
    gene_DESeq[i,22] <- 'U2OS'
  } else {
    gene_DESeq[i,22] <- 'none'
  }

  if (gene_DESeq[i,17] < 0.05 & gene_DESeq[i,21] < 0.05){
    gene_DESeq[i,23] <- 'both'
  } else if (gene_DESeq[i,17] < 0.05){
    gene_DESeq[i,23] <- 'A549'
  } else if (gene_DESeq[i,21] < 0.05){
    gene_DESeq[i,23] <- 'U2OS'
  } else {
```

```

    gene_DESeq[i,23] <- 'none'
  }
}
gene_DESeq$sig_1nM <- factor(gene_DESeq$sig_1nM,
                             levels = c("none", "U2OS", "A549", "both"))
gene_DESeq$sig_100nM <- factor(gene_DESeq$sig_100nM,
                               levels = c("none", "U2OS", "A549", "both"))

gene_DESeq <- gene_DESeq[order(gene_DESeq$sig_1nM),]

p1 <- ggplot(gene_DESeq, aes(x = log10(baseMean),
                             y = A549_wt_1vs0dex_log2,
                             color = A549_wt_1vs0dex_fdr < 0.05)) +
  geom_point(size = 0.75, stroke = 0) +
  xlim(1.5,4.5) + ylim(-3.75, 7.25) +
  scale_color_manual(values = c('#515151', '#7570b3')) +
  xlab('mean of normalized counts') + ylab('log2 fold-change') +
  ggtitle('A549 1 nM dex') +
  geom_hline(yintercept = 0, linetype = 'dashed') +
  labs(colour = 'sig') + gg_options

p2 <- ggplot(gene_DESeq, aes(x = log10(baseMean),
                             y = U2OS_wt_1vs0dex_log2,
                             color = U2OS_wt_1vs0dex_fdr < 0.05)) +
  geom_point(size = 0.75, stroke = 0) +
  xlim(1.5,4.5) + ylim(-3.75, 7.25) +
  scale_color_manual(values = c('#515151', '#1b9e77')) +
  xlab('mean of normalized counts') + ylab('log2 fold-change') +
  ggtitle('U2OS 1 nM dex') +
  geom_hline(yintercept = 0, linetype = 'dashed') +
  labs(colour = "sig") + gg_options

gene_DESeq <- gene_DESeq[order(gene_DESeq$sig_100nM),]

p3 <- ggplot(gene_DESeq, aes(x = log10(baseMean),
                             y = A549_wt_100vs0dex_log2,
                             color = A549_wt_100vs0dex_fdr < 0.05)) +
  geom_point(size = 0.75, stroke = 0) +
  xlim(1.5,4.5) + ylim(-3.75, 7.25) +
  scale_color_manual(values = c('#515151', '#7570b3')) +
  xlab('mean of normalized counts') + ylab('log2 fold-change') +
  ggtitle('A549 100 nM dex') +
  geom_hline(yintercept = 0, linetype = 'dashed') +
  labs(colour = 'sig') + gg_options

p4 <- ggplot(gene_DESeq, aes(x = log10(baseMean),
                             y = U2OS_wt_100vs0dex_log2,
                             color = U2OS_wt_100vs0dex_fdr < 0.05)) +
  geom_point(size = 0.75, stroke = 0) +
  xlim(1.5,4.5) + ylim(-3.75, 7.25) +
  scale_color_manual(values = c('#515151', '#1b9e77')) +
  xlab('mean of normalized counts') + ylab('log2 fold-change') +
  ggtitle('U2OS 100 nM dex') +

```



```

geom_hline(yintercept = 0, linetype = 'dashed') +
labs(colour = "sig") + gg_options

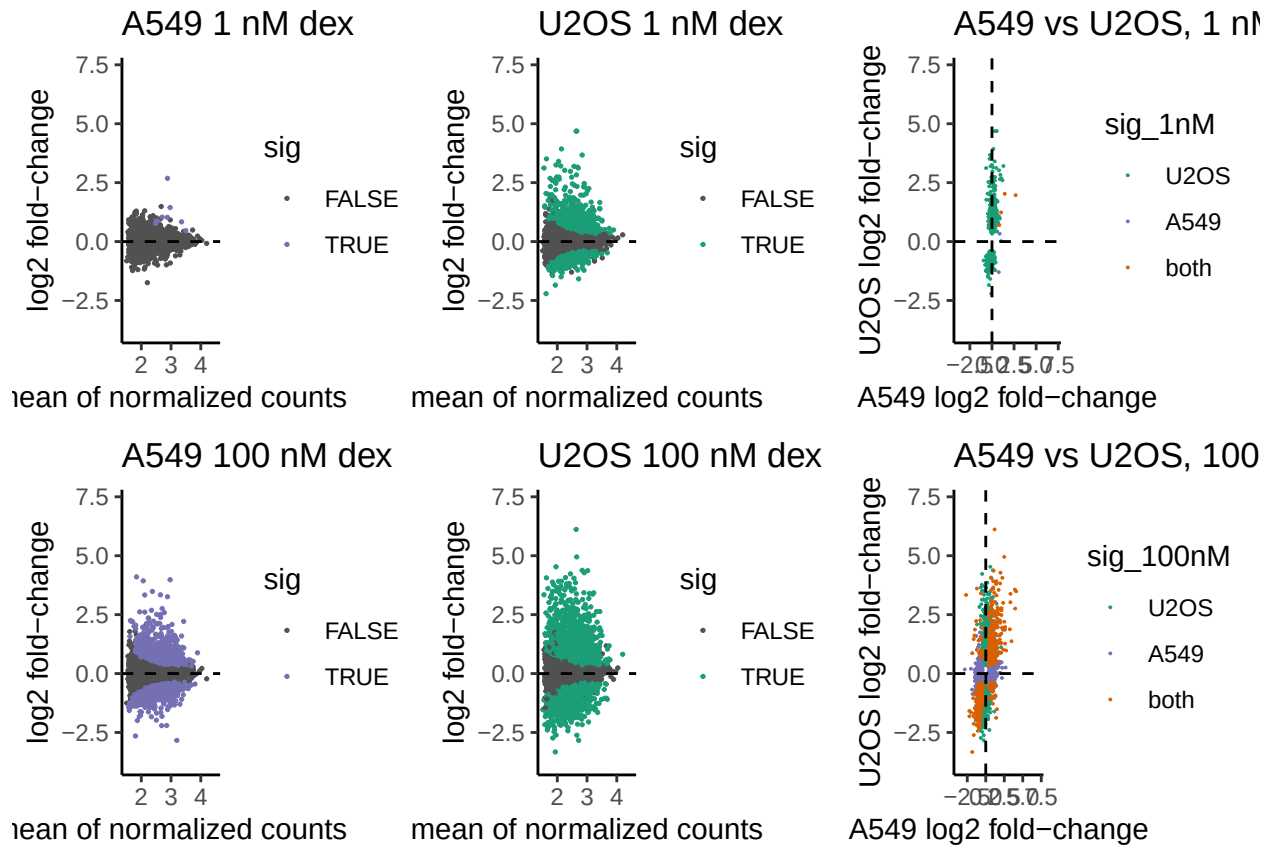
sig1 <- gene_DESeq[gene_DESeq$sig_1nM != 'none',]

p5 <- ggplot(sig1, aes(x = A549_wt_1vs0dex_log2,
y = U2OS_wt_1vs0dex_log2, color = sig_1nM)) +
geom_point(size = 0.5, stroke = 0) +
scale_color_manual(values = c("#1b9e77", "#7570b3", "#d95f02")) +
xlab('A549 log2 fold-change') + ylab('U2OS log2 fold-change') +
ggtitle('A549 vs U2OS, 1 nM dex') +
xlim(-3.75, 7.25) + ylim(-3.75, 7.25) +
geom_vline(xintercept = 0, linetype = 'dashed') +
geom_hline(yintercept = 0, linetype = 'dashed') + gg_options

sig100 <- gene_DESeq[gene_DESeq$sig_100nM != 'none',]
p6 <- ggplot(sig100, aes(x = A549_wt_100vs0dex_log2,
y = U2OS_wt_100vs0dex_log2, color = sig_100nM)) +
geom_point(size = 0.5, stroke = 0) +
scale_color_manual(values = c("#1b9e77", "#7570b3", "#d95f02")) +
xlab('A549 log2 fold-change') + ylab('U2OS log2 fold-change') +
ggtitle('A549 vs U2OS, 100 nM dex') +
xlim(-3.75, 7.25) + ylim(-3.75, 7.25) +
geom_vline(xintercept = 0, linetype = 'dashed') +
geom_hline(yintercept = 0, linetype = 'dashed') + gg_options

p7 <- grid.arrange(p1, p2, p5, p3, p4, p6, nrow = 2)

```



```
cor.test(gene_DESeq$A549_wt_1vs0dex_log2,
         gene_DESeq$U2OS_wt_1vs0dex_log2,
         method = 'spearman')
```

```
##
## Spearman's rank correlation rho
##
## data: gene_DESeq$A549_wt_1vs0dex_log2 and gene_DESeq$U2OS_wt_1vs0dex_log2
## S = 9.9469e+10, p-value < 2.2e-16
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##      rho
## 0.3770218
```

```
cor.test(gene_DESeq$A549_wt_100vs0dex_log2,
         gene_DESeq$U2OS_wt_100vs0dex_log2,
         method = 'spearman')
```

```
##
## Spearman's rank correlation rho
##
## data: gene_DESeq$A549_wt_100vs0dex_log2 and gene_DESeq$U2OS_wt_100vs0dex_log2
## S = 1.0297e+11, p-value < 2.2e-16
## alternative hypothesis: true rho is not equal to 0
```

```
## sample estimates:
##      rho
## 0.355088
```

```
print (c('Number of significantly upregulated genes in A549 after 1 nM dex',
        nrow(gene_DESeq[gene_DESeq$A549_wt_1vs0dex_fdr<0.05 &
                        gene_DESeq$A549_wt_1vs0dex_log2>0,
                        ])))
```

```
## [1] "Number of significantly upregulated genes in A549 after 1 nM dex"
## [2] "8"
```

```
print (c('Number of significantly downregulated genes in A549 after 1 nM dex',
        nrow(gene_DESeq[gene_DESeq$A549_wt_1vs0dex_fdr<0.05 &
                        gene_DESeq$A549_wt_1vs0dex_log2<0,
                        ])))
```

```
## [1] "Number of significantly downregulated genes in A549 after 1 nM dex"
## [2] "0"
```

```
print (c('Number of significantly upregulated genes in A549 after 100 nM dex',
        nrow(gene_DESeq[gene_DESeq$A549_wt_100vs0dex_fdr<0.05 &
                        gene_DESeq$A549_wt_100vs0dex_log2>0,])))
```

```
## [1] "Number of significantly upregulated genes in A549 after 100 nM dex"
## [2] "480"
```

```
print (c('Number of significantly downregulated genes in A549 after 100 nM dex',
        nrow(gene_DESeq[gene_DESeq$A549_wt_100vs0dex_fdr<0.05 &
                        gene_DESeq$A549_wt_100vs0dex_log2<0,])))
```

```
## [1] "Number of significantly downregulated genes in A549 after 100 nM dex"
## [2] "352"
```

```
print (c('Number of significantly upregulated genes in U2OS after 1 nM dex',
        nrow(gene_DESeq[gene_DESeq$U2OS_wt_1vs0dex_fdr<0.05 &
                        gene_DESeq$U2OS_wt_1vs0dex_log2>0,
                        ])))
```

```
## [1] "Number of significantly upregulated genes in U2OS after 1 nM dex"
## [2] "401"
```

```
print (c('Number of significantly downregulated genes in U2OS after 1 nM dex',
        nrow(gene_DESeq[gene_DESeq$U2OS_wt_1vs0dex_fdr<0.05 &
                        gene_DESeq$U2OS_wt_1vs0dex_log2<0,
                        ])))
```

```
## [1] "Number of significantly downregulated genes in U2OS after 1 nM dex"
## [2] "237"
```

```
print (c('Number of significantly upregulated genes in U2OS after 100 nM dex',
        nrow(gene_DESeq[gene_DESeq$U2OS_wt_100vs0dex_fdr<0.05 &
                        gene_DESeq$U2OS_wt_100vs0dex_log2>0,])))
```

```
## [1] "Number of significantly upregulated genes in U2OS after 100 nM dex"
## [2] "1195"
```

```
print (c('Number of significantly downregulated genes in U2OS after 100 nM dex',
        nrow(gene_DESeq[gene_DESeq$U2OS_wt_100vs0dex_fdr<0.05 &
                        gene_DESeq$U2OS_wt_100vs0dex_log2<0,])))
```

```
## [1] "Number of significantly downregulated genes in U2OS after 100 nM dex"
## [2] "1114"
```

##Comparing 1 nM to 100 nM treatment per cell type

```
gene_DESeq$A549_sig <- ''
gene_DESeq$U2OS_sig <- ''

for (i in 1:nrow(gene_DESeq)){
  if (gene_DESeq[i,15] < 0.05 & gene_DESeq[i,17] < 0.05){
    gene_DESeq[i,24] <- 'both'
  } else if (gene_DESeq[i, 15] < 0.05){
    gene_DESeq[i,24] <- '1nM'
  } else if (gene_DESeq[i, 17] < 0.05){
    gene_DESeq[i,24] <- '100nM'
  } else{
    gene_DESeq[i,24] <- 'none'
  }

  if (gene_DESeq[i,19] < 0.05 & gene_DESeq[i,21] < 0.05){
    gene_DESeq[i,25] <- 'both'
  } else if (gene_DESeq[i, 19] < 0.05){
    gene_DESeq[i,25] <- '1nM'
  } else if (gene_DESeq[i, 21] < 0.05){
    gene_DESeq[i,25] <- '100nM'
  } else{
    gene_DESeq[i,25] <- 'none'
  }
}

gene_DESeq$A549_sig <- factor(gene_DESeq$A549_sig,
                             levels = c("none", "1nM", "100nM", "both"))
gene_DESeq$U2OS_sig <- factor(gene_DESeq$U2OS_sig,
                             levels = c("none", "1nM", "100nM", "both"))

p8 <- ggplot(gene_DESeq, aes(x = A549_wt_1vs0dex_log2,
                             y = A549_wt_100vs0dex_log2, color = A549_sig)) +
  geom_point(size = 1, stroke = 0) +
```

```

xlab('1 nM log2 fold-change') + ylab('100 nM log2 fold-change') +
ggtitle('A549') + xlim(-3.75, 7.25) + ylim(-3.75, 7.25) +
geom_vline(xintercept = 0, linetype = 'dashed') +
geom_hline(yintercept = 0, linetype = 'dashed') + gg_options +
scale_color_manual(values = c('#515151', '#46ACC8', '#E58601', '#B40F20'))

cor.test(gene_DESeq$A549_wt_1vs0dex_log2, gene_DESeq$A549_wt_100vs0dex_log2, method = 'spearman')

##
## Spearman's rank correlation rho
##
## data: gene_DESeq$A549_wt_1vs0dex_log2 and gene_DESeq$A549_wt_100vs0dex_log2
## S = 7.9374e+10, p-value < 2.2e-16
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##      rho
## 0.5028757

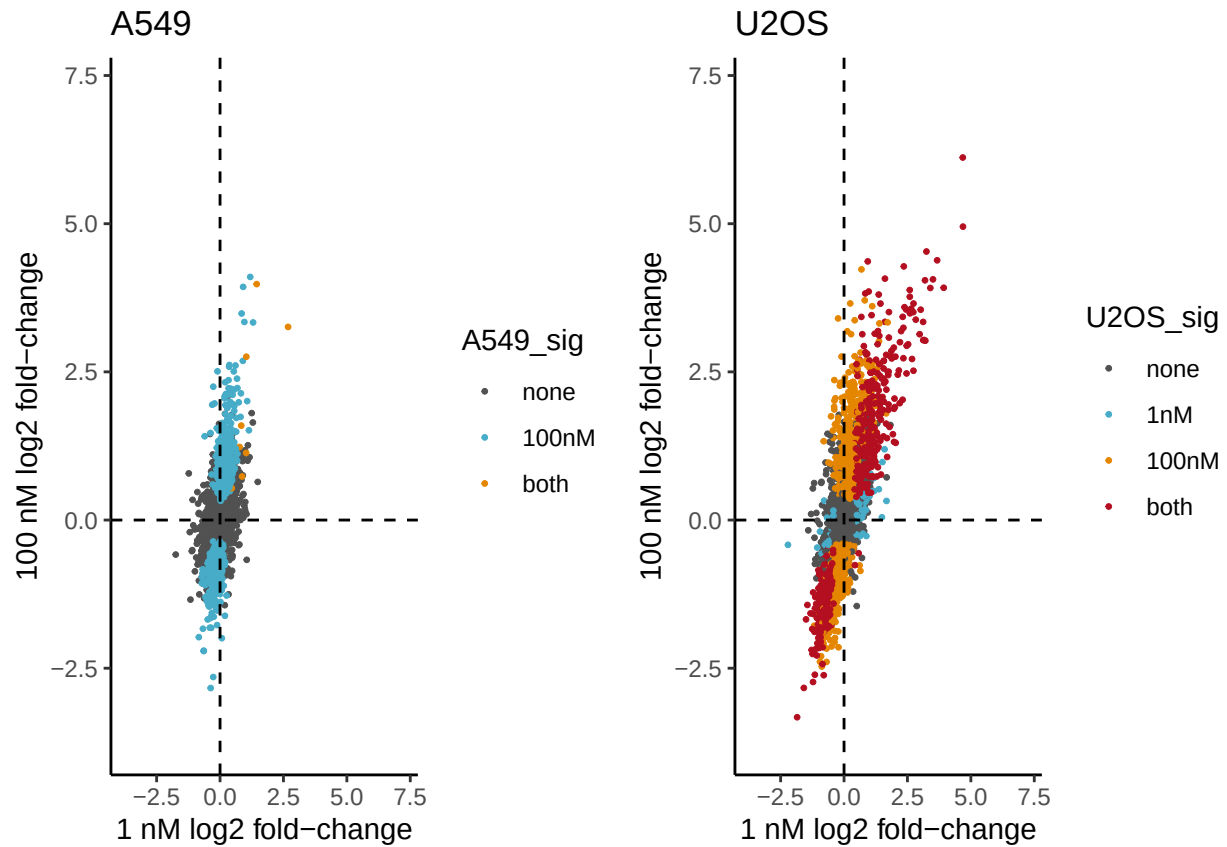
p9 <- ggplot(gene_DESeq, aes(x = U2OS_wt_1vs0dex_log2,
y = U2OS_wt_100vs0dex_log2, color = U2OS_sig)) +
geom_point(size = 1, stroke = 0) +
xlab('1 nM log2 fold-change') + ylab('100 nM log2 fold-change') +
ggtitle('U2OS') + xlim(-3.75, 7.25) + ylim(-3.75, 7.25) +
geom_vline(xintercept = 0, linetype = 'dashed') +
geom_hline(yintercept = 0, linetype = 'dashed') + gg_options +
scale_color_manual(values = c('#515151', '#46ACC8', '#E58601', '#B40F20'))

cor.test(gene_DESeq$U2OS_wt_1vs0dex_log2, gene_DESeq$U2OS_wt_100vs0dex_log2, method = 'spearman')

##
## Spearman's rank correlation rho
##
## data: gene_DESeq$U2OS_wt_1vs0dex_log2 and gene_DESeq$U2OS_wt_100vs0dex_log2
## S = 4.381e+10, p-value < 2.2e-16
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##      rho
## 0.7256159

p10 <- grid.arrange(p8, p9, nrow = 1)

```



##Heatmap of significantly changed genes

```
sig <- unique(rbind(sig1, sig100))

sig <- sig[,14:21]
sig$category <- ''

for (i in 1:nrow(sig)){
  if (sig[i,4] < 0.05 & sig[i,8]<0.05){
    if (sig[i,3] > 0 & sig[i,7] > 0){
      sig[i,9] <- 'both_up'
    }
    else if (sig[i,3] < 0 & sig[i,7] < 0){
      sig[i,9] <- 'both_down'
    }
    else{
      sig[i,9] <- 'opposite'
    }
  }
  else if (sig[i,4] <0.05){
    if (sig[i,3] > 0) {
      sig[i,9] <- 'A549_up'
    }
    else {sig[i,9] <- 'A549_down'}
  }
  else{
```

```

    if (sig[i,7] > 0){
      sig[i,9] <- 'U20S_up'
    }
    else{sig[i,9] <- 'U20S_down'}
  }
}
sig$category <- factor(sig$category, levels =
  c('both_up' , 'A549_up' , 'U20S_up' ,
    'both_down' , 'A549_down' , 'U20S_down', 'opposite'))
table(sig$category)

```

```

##
##   both_up   A549_up   U20S_up both_down A549_down U20S_down  opposite
##       262       186       940       185       141       912        58

```

```

sig <- sig[order(sig$category),]

pheatmap(sig[,c(1,3,5,7)],
  show_rownames = F,
  color = rev(colorRampPalette(brewer.pal(n = 11, name = "RdBu")) (99)),
  breaks = c(seq(-3.0, -2, length.out = 20),
    seq(-1.9, -.5, length.out = 20),
    seq(-0.4, 0.4, length.out = 20),
    seq(0.5, 1.9, length.out = 20),
    seq(2, 6, length.out = 20)),
  cluster_rows = F, cluster_cols = F)

```

